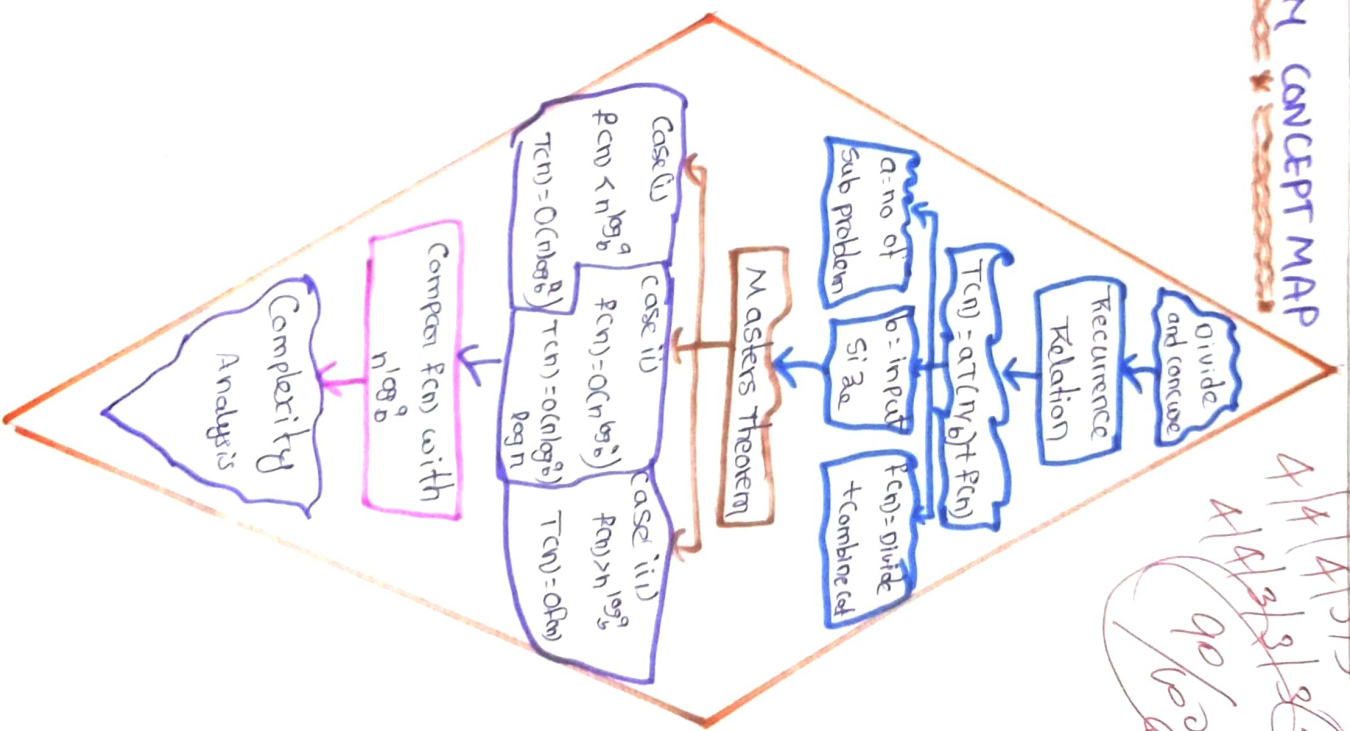


DIVIDE AND CONQUER - MASTER THEOREN CONCEPT MAP



CONCEPT MAP

2. ALGORITHM

- A finite set of instructions
- solves the problem
- Running time depends on input size (n)



3. GROWTH FUNCTION

- Describes how running time ($T(n)$) increases
- Use to Analyse Algorithm Efficiency

3. BIG-O - NOTATION

- Provides as asymptotic upper bound
- $T(n) \leq c \cdot f(n)$ for $n \geq n_0$
- Represents maximum growth

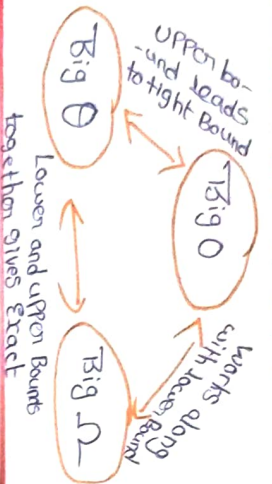
4-BIG-THETA NOTATION

- Provides an asymptotically tight Bound
- $C_1 \cdot f(n) \leq T(n) \leq C_2 \cdot f(n)$ for $n \geq n_0$

5. BIG-OMEGA NOTATION

- Lower Bound
- $T(n) \geq c \cdot f(n)$ for $n \geq n_0$
- Focus on the best case growth

Relation $Big O, \Omega, \Theta$



6. TIGHT BOUND when upper and Lower Bounds meet
 $T(n) O(f(n)) = \Omega(f(n))$, then $O(f(n))$
 upper Bound = Lower Bound $\Rightarrow$ Tight Bounds

7. EXACT ORDER OF GROWTH

- when $T(n) = O(f(n))$, we know the exact asymptotic growth. It precisely running time with n

NOTATION

BIG O

Tells it most how fast it grows

BIG-OMEGA

Tells "at least" how fast it grows

BIG-THETA

Tells "exactly" how fast it grows

therefore gives precise complexity while O and Ω give approx